Process Characterization Plan

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCP-005

Process Development / Manufacturing Science & Technology (MSAT)

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; CQA, critical quality attribute; CV, column volume; DNA, deoxyribonucleic acid; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; Fc, fragment crystallizable region of an immunoglobulin; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG, immunoglobulin G; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PPQ, process performance qualification; PRESS, prediction error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Protein A chromatography is the first chromatographic step of the A-Mab purification train. This plan defines the Stage-1 characterization study for that step, and it fixes the acceptance and decision criteria that the executed study will be judged against (U.S. Food and Drug Administration 2011). It is written and approved before any characterization run is executed, and the study itself will be performed in a qualified scale-down model at the sending site (Cambridge) and reported in PCR-005.

Table 1.1: The A-Mab drug-substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The step takes clarified harvest from Step 4, binds the antibody on an immobilized Protein A resin, and elutes it at low pH into the viral inactivation hold of Step 6. It sets one quality attribute, leached Protein A, and it is the principal point of host cell protein and residual DNA removal in the process (Table 1.1). Viral clearance can be shown across affinity capture steps of this kind, but no viral clearance claim will be made for this one. The cumulative claim rests on Steps 6, 8 and 9 and is consolidated in PCMR-001 (International Council for Harmonisation 2023a).

This plan covers the 4 process parameters that RA-001 assigned to multivariate study and the 2 parameters it assigned to univariate study, over the characterization ranges given in §6.1. Three responses will be measured on every run (pool host cell protein, leached Protein A and step yield), while residual DNA and aggregate will be measured on the reduced scheme of §5.4.

Three areas that a reader might expect here are outside the scope of this plan and are covered by other documents of the characterization package. Resin packing, sanitization, re-use and lifetime are governed by SOP-2008 and are characterized in the resin life-cycle study referenced there. Analytical method performance is established by the validations listed in Table 3.2 and is not re-established here. Hold times between unit operations, and the equivalence of the receiving site's equipment, are covered by PTP-001.

2 Objectives

The study pursues the objectives below, and each of them is answered by a named section of PCR-005.

- (i) Quantify the effect of each studied parameter on pool host cell protein, on leached Protein A and on step yield, across the characterization ranges of §6.1.
- (ii) Identify the interactions between the multivariate parameters, and estimate the curvature of any response that shows it.
- (iii) Define a multivariate operating region for the parameters that govern a quality attribute, and determine a proven acceptable range for each governed attribute and parameter combination.
- (iv) Provide the effect estimates and the operating-region risk assessment needed to classify every parameter of the step under SOP-4001.
- (v) Justify the normal operating ranges to be transferred to the receiving site, and state the contribution this step makes to the overall control strategy.

The enhanced approach of ICH Q8(R2) and ICH Q11 is followed throughout, criticality is treated as a continuum in the sense of ICH Q9(R1), and both the methodology and the attribute framework are those of the A-Mab case study (International Council for Harmonisation 2009, 2012, 2023b; CMC Biotech Working Group 2009).

3 Related documents

Table 3.1 lists the documents this plan depends on (PTP-001, RA-001 and PCMP-001) and the documents it feeds (PCR-005 and PCMR-001). The controlled procedures and validated analytical methods that govern the study are listed in Table 3.2, and all identifiers in both tables are placeholders in a synthetic corpus.

Table 3.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-005	Process Characterization Report (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2: Controlled procedures and validated analytical methods for the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The capture step uses an immobilized Protein A resin operated in bind and elute mode. The clarified harvest is loaded onto an equilibrated column and the antibody binds through its Fc region, while host cell protein, DNA and media components pass into the flow through, and a wash then displaces the weakly retained impurities from the bed. The bound antibody is eluted with a low pH buffer, and the eluate pool is collected until a defined column volume (CV) has passed. The pool feeds the low pH hold of Step 6 without adjustment, so the elution buffer pH also sets the starting condition of that step.

Affinity capture on this platform is well established. The resin, the buffer system and the operating sequence are those used for three previous humanized IgG1 products (X-Mab, Y-Mab and Z-Mab) and for the A-Mab clinical campaigns. That history shows the step to be robust and its performance to be governed by a small number of parameters. Characterization therefore confirms and bounds behaviour that is already understood, and the study is designed on that basis.

Three mechanisms frame what the study expects to find. Protein load determines how much antibody competes for the available binding sites, so a high load is expected to raise the host cell protein carried into the eluate, and at short residence times it is expected to lose product to breakthrough. Elution buffer pH sets the strength of the elution and, through it, the balance between recovery of the antibody and carry-over of impurity. A lower pH releases the antibody more completely, and it also mobilizes impurities that remain associated with the resin and accelerates hydrolysis of the ligand, so both pool host cell protein and leached Protein A are expected to rise as the elution pH falls. The end of pool collect determines how much of the trailing edge of the elution peak is taken, and the trailing edge is where the later eluting impurities (host cell protein and leached ligand) are concentrated.

Load flow rate acts through a different route from the other three factors. It changes the residence time on the column and hence the dynamic binding capacity, so its effect is expected on step yield and not on the quality attributes. The executed study will confirm or refute each of these expectations, and PCR-005 will report the mechanistic interpretation alongside the effect estimates.

4.2 Quality attributes in scope

The step sets one of the attributes listed in Table 4.1, clears two of them, and monitors the fourth.

Table 4.1: Quality attributes governed or monitored at the capture step, with their drug-substance acceptance criteria.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0–5 ppm	L-M	16
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60

Leached Protein A is formed here and nowhere else in the process, so the criterion of 5 ppm is the direct responsibility of this study. Host cell protein enters with the clarified harvest and this step is the principal point of its removal, but the drug-substance criterion of 100 ng/mg is met only after the two polishing steps, which are characterized in PCP-007 and PCP-008. Residual DNA carries the lowest criticality of the four attributes and is cleared across three steps.

Aggregate is formed in the production bioreactor and reduced at cation exchange (Table 1.1). It is monitored across the capture step for one reason. Elution at low pH is the point in the process at which acid induced aggregation would first appear, and the lower edge of the elution pH range studied here is set by that risk (§6.1). No aggregate reduction is claimed for this step.

The glycan and charge-variant attributes are not in scope here, because they are formed in the production bioreactor and platform experience shows that affinity capture under these conditions does not discriminate between the variants concerned (PCR-003 reports their characterization).

4.3 Risk-based prioritization of parameters

RA-001 assigned the study type of every parameter of the step before this plan was written, and that assignment is reproduced in the last column of Table 6.1. The ranking gave each parameter a score for its potential main effect on a quality attribute and a second score for its potential to interact with other parameters, and the product of the two determined the study type (CMC Biotech Working Group 2009). Parameters that scored high on both went to the multivariate design, and parameters whose effect is expected to be monotonic and independent of the others went to univariate assessment.

No parameter of this step was excluded from study, which is a consequence of the small parameter set, and the knowledge space of the step is therefore the full cross product of the characterization ranges in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Every characterization run will be performed on a bench-scale column that is a qualified model of the commercial step. The model is designed on the established principles of chromatographic scaling, and the intent behind it is that the product sees the same conditions at both scales. It holds the bed height, the linear flow velocity, the protein load per litre of resin, and the wash and elution volumes expressed in column volumes equal to their commercial values, so that the residence time and the number of column volumes seen by the product are preserved. Column efficiency will be matched on plate count and peak asymmetry, which are the two measures that detect a poorly packed bed.

Qualification will follow SOP-1001 and will precede the first characterization run, and in it triplicate scale-down runs at the set-point condition will be compared with the corresponding at-scale data for step yield, pool host cell protein, leached Protein A and residual DNA. The model will be accepted when no statistically significant difference is found between scales in those performance attributes and when column efficiency falls inside the range recorded for the commercial column. The qualification record will be filed under SOP-1001 and summarized in the materials and methods section of PCR-005.

The centre points of the two designs serve a second purpose beyond their role in the designs themselves. They are true replicates of one condition, so their variance estimates the pure error of the whole measurement system, including the column, the operation and the assay. That estimate is the reference against which the lack of fit of each model is tested (§5.5).

The warrant the qualified model provides is bounded. It supports claims about operating ranges, about impurity clearance and about the shape of the response surfaces. It does not support claims about packing a column at commercial diameter, about resin lifetime, or about the hold time between the eluate pool and the low pH inactivation step. Those are addressed by SOP-2008 and PTP-001, and the first of them is confirmed at commercial scale during PPQ.

5.2 Operation

The load material will be clarified harvest produced under the conditions characterized in PCR-004 and released against the platform input specification. A single pooled lot will supply every run of a given design, so that variation in the feed cannot be confounded with the studied factors. The input attributes of that pool (titre, host cell protein, residual DNA and aggregate) will be measured once and reported with the study.

Each run will follow SOP-2007, under which the column is equilibrated, loaded to the assigned protein load at the assigned flow rate, washed, and eluted with the elution buffer at its assigned pH. Pool collection starts at a fixed absorbance threshold on the ascending edge of the elution peak and stops at the assigned column volume, and the column is then regenerated and sanitized under SOP-2008.

Two inputs are controlled to platform specification and are not characterized here. The resin lot and the composition of the equilibration, wash and elution buffers are fixed for the whole study, and their identity and quality are verified on receipt.

5.3 Analytical methods

Every response will be measured by a validated method, and Table 5.1 lists them with the validation that governs each one. Leached Protein A is measured by ELISA under AMV-3016, and host cell protein by the platform ELISA under AMV-3012. Residual DNA is measured by qPCR under AMV-3014 and aggregate by SEC-HPLC under AMV-3011, while step yield is calculated from the protein concentration and the recorded volume of the load and of the eluate pool.

Table 5.1: Validated analytical methods supporting the study.

Reference	Title	Type
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Assay precision is most consequential for leached Protein A, because eluate pool concentrations fall in the lower part of the working range of the ELISA. The reportable value, the calibration range and the intermediate precision of each method are given in the validation reports referenced in Table 5.1. Where an estimated effect is of the same order as the precision of the assay that measured it, PCR-005 will report it as such and will not derive an operating limit from it.

5.4 Sampling plan

Samples will be taken from the load, the flow through, the final wash and the eluate pool of every run. Pool host cell protein, leached Protein A and protein concentration will be measured on every pool, since all three are responses of the designs. Residual DNA and aggregate will be measured on the load and on the centre-point pools of both designs, which supports a clearance factor for the step without carrying either attribute as a modelled response.

Centre-point replication differs between the two designs, since the screening design carries 3 centre points and the response-surface design carries 4. Run order will be randomized within each design, and the samples of a design will be assayed in a single analytical campaign, so that analytical drift is not aliased with a factor.

5.5 Statistical methods

The analysis will be performed under SOP-1002, and each factor is coded so that its set-point is zero and the edges of its characterization range are minus one and plus one. Coding places the factors on a common scale and makes the estimated coefficients comparable across parameters measured in different units (protein load in g/L resin, pH, linear velocity and column volumes).

The screening data will be fitted by ordinary least squares to a model carrying all main effects and all two-factor interactions. An effect will be reported as twice the coded coefficient, which is the change in the response across the full characterization range of that factor. Significance will be judged at alpha = 0.05, and effect estimates will be reported with their standard errors.

The response-surface data will be fitted to the full quadratic model in the same factors. Adequacy will be judged by the coefficient of determination, by its adjusted and predicted forms, by the overall F test, and by an analysis of variance that partitions the residual into lack of fit and pure error. The predicted coefficient is computed from the leave one out residuals (PRESS), so a model that describes its own data without predicting new data will be visible in the comparison of the two.

Commercial-scale capability will be estimated by Monte-Carlo simulation of 2,000 batches, in which each parameter is drawn inside its normal operating range, the fitted response-surface model is evaluated at every draw, and a one-sided capability index is computed against the acceptance criterion of each attribute in Table 4.1. The simulation propagates the fitted model and the residual variance of the study, and it does not carry the additional variability of commercial-scale operation.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives every parameter of the step, with its set-point, the range over which it will be studied, its normal operating range and the type of study assigned to it.

Table 6.1: Parameters of the capture step, with set-point, characterization range, normal operating range and the study type assigned by RA-001.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	30	10–50	20–40	multivariate
Elution buffer pH	pH	3.55	3.2–3.9	3.4–3.7	multivariate
Load flow rate	cm/hr	200	100–300	150–250	multivariate
End of pool collect	CV	2.6	2–3.2	2.3–2.9	multivariate
Operating temperature	°C	22	15–30	18–26	univariate
Bed height	cm	20	10–30	15–25	univariate

Each characterization range extends beyond its normal operating range on both sides, so that the study covers credible excursions as well as routine operation, and so that the resulting knowledge space is larger than the region the process will be operated in (International Council for Harmonisation 2009). The 4 multivariate factors carry the letter codes of Table 6.2 throughout the design matrices in the appendices.

Table 6.2: Letter codes for the multivariate factors, used in Appendix A and Appendix B.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

The edges of each range are chosen for a stated reason. Protein load will be studied over 10–50 g/L resin. The lower edge represents an early, low-titre harvest, and the upper edge approaches the dynamic binding capacity of the resin at the shortest residence time in the design, which is where breakthrough would first appear. Elution buffer pH will be studied over 3.2–3.9. Its lower edge is bounded by the risk of acid induced aggregation, which platform experience places below the value studied here, and its upper edge is bounded by incomplete elution and the yield loss that accompanies it. Load flow rate will be studied over 100–300 cm/hr. That range spans the residence times achievable at commercial scale within the packing tolerance of the column. End of pool collect will be studied over 2–3.2 CV. Its lower edge truncates the peak before the trailing edge, and its upper edge extends well into it.

The two univariate parameters are ranged on the same principle, so operating temperature will be studied over 15–30 °C, which covers the extremes of the chromatography suite, and bed height over 10–30 cm, which covers the packing tolerance of the commercial column.

6.2 Screening design

The screening stage establishes which factors matter. A two-level full factorial in the 4 multivariate factors will be run and augmented with 3 centre points, giving 19 runs in total. The design is not fractional, so all 4 main effects and all 6 two-factor interactions are estimated without aliasing.

A design of this size identifies effects. It does not by itself support prediction, because it carries no quadratic terms and few residual degrees of freedom once the interactions are fitted. The predictive model of this study is the response-surface model of §6.3, and the operating region and the proven acceptable ranges will be derived from that model and not from the screening fit.

Run order will be randomized, and the centre points will be distributed through the sequence, so that drift in the column or in the assay appears as centre-point variation and is separated from the factor effects. The planned matrix is given in Appendix A.

6.3 Response-surface design

A face-centred central composite design will follow in the same 4 factors, and it comprises 16 factorial runs, 8 axial runs and 4 centre points, giving 28 runs. The axial points sit on the faces of the cube, so no run is set outside the characterization range of any factor, and the design stays inside the region the scale-down model was qualified over.

The quadratic terms are the reason for this second design. Curvature distinguishes a response with an interior optimum from one that changes monotonically across the range, and only the second kind can be summarized honestly by a univariate limit.

The fitted quadratic model is the model to which the criteria of §7 and the analysis of §8 will be applied.

Figure 6.1 shows the planned design in the plane of protein load and elution buffer pH, which are the two parameters expected to govern the quality attributes of the step, and it also shows that the normal operating range sits well inside the characterized region on both axes.

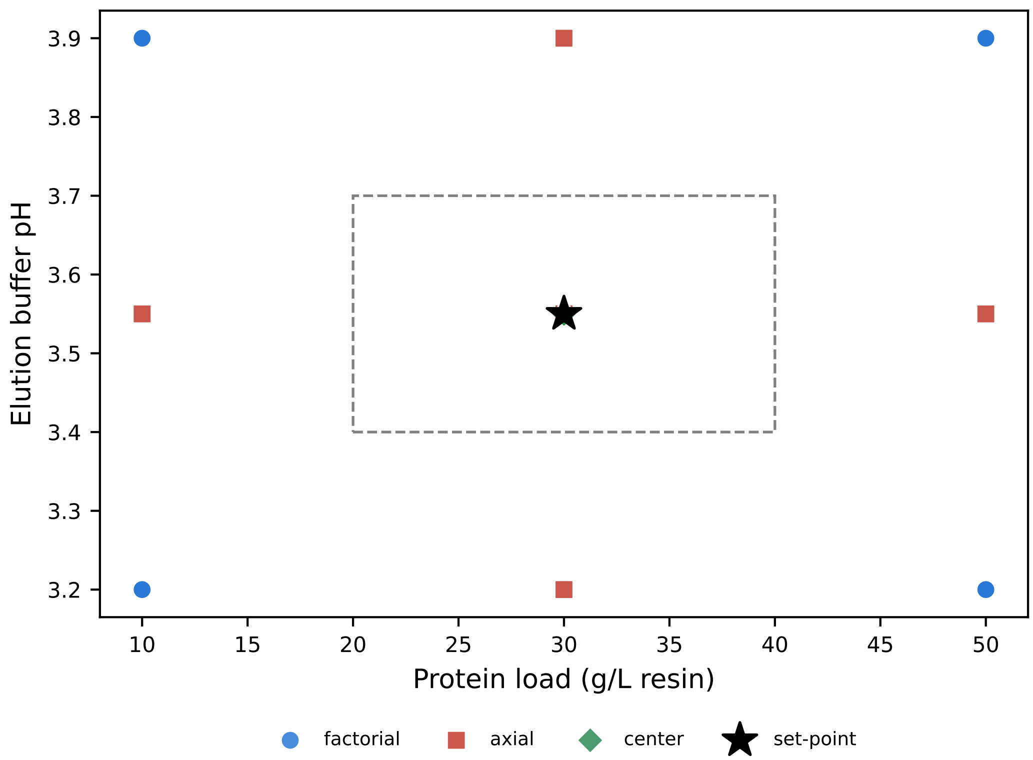

Figure 6.1: Planned response-surface design, projected onto the plane of protein load and elution buffer pH. Runs coincide in this projection because the other two factors are not shown. The dashed box is the normal operating range and the star marks the set-point.

The planned matrix is given in Appendix B.

6.4 Univariate assessment

Operating temperature and bed height will be assessed one factor at a time, with every other parameter held at its set-point. RA-001 kept both out of the multivariate design for a mechanistic reason. Temperature acts on binding kinetics and on the rate of ligand hydrolysis, and both processes are slow compared with the residence time of the step. Bed height acts on residence time at a fixed linear velocity, so its influence is already spanned by the load flow rate factor of the multivariate design.

Each of the two will be run at both edges of its characterization range and at its set-point, and the responses measured are those of Table 5.1. The outcome will be reported as a proven acceptable range for the parameter, on the same basis as §8, and not as a term in the response-surface model.

7 Acceptance and decision criteria

The criteria in this section are fixed before the study is executed, and PCR-005 will judge the study against them. They are stated as rules, so that the disposition of a result does not depend on the result.

The scale-down model must be qualified before the first characterization run, against the criterion of §5.1. If the model fails to qualify, the study stops, the model is modified, and it is requalified before any design is executed.

A fitted model will be accepted as adequate when its lack of fit is not significant against pure error at alpha = 0.05 and when its adjusted coefficient of determination supports the effects being reported. A model will be described as predictive only when its predicted coefficient of determination is close to its adjusted value. Where the two diverge, the model will be reported as descriptive over the ranges studied, and no prediction outside the observed conditions will be made from it.

A response with no significant term is a result and will be reported as one. That response will be described as robust over the characterization ranges studied. Its parameters will remain in the knowledge space as the evidence for that robustness, and no operating limit will be derived from an effect that the study could not resolve.

Attribute acceptance is taken from the drug-substance criteria in Table 4.1, so leached Protein A in the eluate pool must not exceed 5 ppm. Pool host cell protein will be judged against the drug-substance criterion of 100 ng/mg, which is a conservative reference for an intermediate pool, because cation exchange and anion exchange each remove further host cell protein (PCP-007, PCP-008). Where that criterion is not met across the characterization range, the operating limits for host cell protein will be set jointly with the polishing steps and consolidated in PCMR-001, and this report will state the pool levels the polishing steps must accept.

Step yield carries no acceptance criterion of that kind. It is a process-performance attribute, and it will be reported with its model and used to justify the normal operating ranges of the parameters that affect it.

The operating region will be declared as the multivariate subset of the characterization ranges over which every attribute with a criterion is predicted to stay inside that criterion. The normal operating ranges of Table 6.1 must lie inside that region with margin on every axis. If they do not, PCR-005 will narrow the normal operating range and carry the change into the control strategy, and the acceptance criterion will not be relaxed to accommodate the observed performance.

Capability will be judged against the minimum capability index defined in PCMP-001, and it will be reported for each attribute together with its margin to the limit.

Parameter classification will follow the decision logic of SOP-4001 and will be recorded in PCR-005. No classification is asserted in this plan.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of one parameter over which a quality attribute stays within acceptance, and it complements the multivariate operating region of §7. The study will report both, and PCR-005 will state which of the two binds the operating limits of the step.

The acceptance basis is fixed first. For each response that maps to a quality attribute, the criterion is the drug-substance specification given in Table 4.1. No back calculation from a cumulative requirement is needed at this step, because that calculation applies only to a viral clearance attribute, whose step-level requirement is the cumulative requirement less the clearance credited to the other steps, and this step makes no viral clearance claim.

Two analyses will be run for every combination of a governed attribute and a multivariate parameter. The first holds the remaining parameters at their set-points and scans the parameter of interest across its characterization range. The second varies the remaining parameters inside their normal operating ranges, by Monte-Carlo of the fitted response-surface model with 2,000 draws at each of 81 grid points, and takes the 95 % predictive interval of the attribute. The first analysis answers what the parameter does on its own. The second answers whether the range survives normal co-variation of the rest of the operating point, and it is the narrower of the two.

The decision rule is the same in both analyses. The proven acceptable range is the contiguous interval containing the set-point over which the attribute stays within its acceptance criterion. Where that interval covers the whole characterization range, the parameter is proven acceptable across everything studied, and the binding constraint on the operating region is then the multivariate corner and not the univariate range. Where the interval is narrower, PCR-005 will name the attribute that binds it and give the remaining margin to the normal operating range. Where the set-point condition itself does not meet the criterion, no proven acceptable range exists for that combination, and the report will say so plainly and refer the attribute to the downstream steps.

Each governed attribute will be reported with a two-panel figure for its governing parameter, in which the left panel is the analysis at set-point and the right panel the analysis with the other parameters varying inside their normal operating ranges. Both panels carry the acceptance limit, the shaded acceptable region, the set-point and the normal operating range. The full table of ranges will cover every combination of a governed attribute and a multivariate parameter, in both analyses.

9 Data management and integrity

Raw data will be captured in the electronic laboratory notebook and the chromatography data system at the point of generation, the audit trail of each system will remain enabled for the duration of the study, and the records will meet ALCOA+ expectations. Analytical results will be transferred into the statistical analysis by a validated export, and the analysis script and its input data will be archived together under SOP-1002. A second qualified person will review every transcription and every fitted model before its results are used in PCR-005.

10 Roles and responsibilities

Table 10.1 assigns the work of the study, and the author of this plan is accountable for its execution as written. Any departure from the plan is recorded as a deviation in PCR-005.

Table 10.1: Roles and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owns the plan; qualifies the scale-down model; executes the runs; authors PCR-005
Analytical Development	Performs the assays under the validations in Table 5.1; reports method performance
Statistics	Reviews the designs before execution; fits and diagnoses the models; reviews the capability analysis
Manufacturing Science	Confirms commercial-scale equivalence of the operation and reviews the proposed operating ranges
Quality Assurance	Approves the plan and the report; dispositions any deviation

11 Deliverables and schedule

The deliverable of this plan is PCR-005, the characterization report for the capture step. The study runs in three phases. The scale-down model is qualified first, the screening design follows, and the response-surface design and the univariate runs are executed last. Each phase is released to the next by a review of its data against §7. No calendar dates are given here, and the campaign schedule is held in PTP-001 and PCMP-001.

12 Risks and assumptions

The study rests on three assumptions, and each is stated here with the consequence that follows if it does not hold, so that PCR-005 can be read against them.

The first assumption is the representativeness of the scale-down model. Qualification under SOP-1001 addresses it directly, but a residual risk remains that packing a column at commercial diameter behaves differently from the bench-scale bed, and that risk is closed at commercial scale during PPQ and not in this study.

The second assumption concerns the use of a single pooled feed lot per design, a choice that removes feed variation from the residual error and is the reason the study cannot detect an interaction between a studied factor and a feed attribute. Feed variability is controlled by the input specification of §5.2 and characterized in PCR-004, and the input attributes of the pool used here will be reported so that the study can be placed against that range.

The third is assay precision at low leached Protein A concentrations (§5.3). If the observed variation of that response is dominated by the assay, the study will not resolve small effects on it, and the response will be reported as robust over the ranges studied instead of being modelled.

One further risk is inherent to the design and is not a failure of it. Conditions at the edges of the characterization ranges may produce material outside an acceptance criterion. That outcome defines an edge of failure and bounds the knowledge space, and it will be reported as such. A change of resin lot during the study would require a bridging run at the set-point condition before the affected design is continued.

13 Approvals

This plan becomes effective when the approvals recorded in Table 1 are complete. Any change after approval is made by a controlled revision of the whole document. A change made after execution has begun is recorded and assessed as a deviation in PCR-005.

14 Appendices A–B — Planned design matrices

Appendix A and Appendix B give the planned design matrices in coded form. The letter codes are those of Table 6.2, and the coded levels of minus one, zero and plus one correspond to the lower edge, the set-point and the upper edge of each characterization range in Table 6.1. No response columns are shown, because no run has been executed.

14.1 Appendix A — Planned screening design matrix

Table 14.1: Planned screening design matrix in coded factors (A, protein load; B, elution buffer pH; C, load flow rate; D, end of pool collect).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

14.2 Appendix B — Planned response-surface design matrix

Table 14.2: Planned face-centred central composite design matrix in coded factors.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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